

HPOS — Factory Operating System

Digital Transformation & Business Management Platform for Himalaya Plast

Transitioning Himalaya Plast from disconnected WhatsApp chats, manual Excel files, and paper QC sheets into a single, unified factory operating system. Complete traceability from raw extrusion materials to dispatched carton boxes.

01. SYSTEM ARCHITECTURE & TECH STACK

BACKEND CORE (70%)

ERPNext v15 + Frappe Engine

The central industrial backbone acting as the **single source of truth** for financial, MRP, inventory, and compliance data.

- **Manufacturing & MRP:** Multi-level BOMs, Work Orders, & Job Cards.
- **Batch Genealogy:** Full backward/forward RM-to-FG tracking.
- **Accounting & Tax:** Invoicing, Payments, P&L, and General Ledger.
- **Statutory GST:** Native e-Invoice (IRN/QR) & E-Way Bill generation.

CUSTOM UI LAYER (30%)

Custom Next.js & React Web Portal

A tailored, high-speed custom frontend communicating with ERPNext via secured REST APIs for seamless factory floor use.

- **Founder Dashboard:** Live P&L, operational runway & risk orders.
- **Visual Planning Board:** Drag-and-drop machine scheduling grid.
- **Shop Floor QR Scanner:** Instant carton & dispatch verification.
- **10-Stage Timeline:** End-to-end Order-to-Cash tracking.

02. CORE BUSINESS WORKFLOWS WE ARE DIGITIZING

OPERATIONAL MODULE	CURRENT PROBLEM / MANUAL GAP	HPOS DIGITAL SOLUTION
Sales & Orders	Quotations and approvals lost across WhatsApp and calls.	Standardized quotes with multi-channel approval audit logs.
Extrusion Planning	Machine scheduling stored in heads, causing bottleneck clashes.	Visual Planning Board assigning Work Orders to specific extruders.
Quality Control	Paper QC sheets; zero digital history or trend analytics.	Digital QC templates (Pin, Width, Leg, Weight, Fit) with 4-state workflow.
Batch Traceability	Days spent manually investigating defective customer batches.	5-second batch genealogy from Delivery Note back to Raw Polymer batch.
Packing & Dispatch	Manual document drafting and missed cartons during truck load.	Carton-wise packing slips with mobile QR scanning verification.
GST & Compliance	Double entry in external software for e-Invoices and E-Way bills.	1-click auto-generation of IRN, QR code, and E-Way bills directly.

03. PROJECT IMPLEMENTATION ROADMAP

Structured agile rollout ensuring your factory operates on a live baseline system within **8 weeks**, followed by custom extensions.

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Phase 0: Setup (W1–3)

VPS Linux environment setup, Dockerized ERPNext configuration, Chart of Accounts, Item Masters & user permission matrix.

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Phase 1: Core MRP (W3–16)

Deploy Order-to-Cash, BOMs, Job Cards, Inventory, QC workflows & India GST Compliance.
Working live core by ~Week 8.

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Phase 2: Custom UI (W16–24)

Custom Next.js interface rollout: Founder Dashboard, 10-Stage Order Tracking Stepper, and Shop-Floor QR Scanning tool.

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Phase 3: Planning (W24–32)

Visual Drag-and-Drop Machine Planning Grid built using live production cycle-time data and capacity benchmarks.

04. WHAT WE NEED FROM YOU

- **Company Masters:** GSTIN, billing/plant addresses, bank details, and Chart of Accounts.
- **Item Catalog & BOMs:** Complete list of raw polymers, masterbatches, profile codes, and standard BOM formulas.
- **Machine & Station List:** Number of extrusion lines, output capacities, and assigned operators.
- **QC Inspection Parameters:** Tolerance bands for Pin Size, Width, Leg Thickness, Weight & Fit tests.
- **Opening Balances:** Active raw material stock, finished goods inventory, and vendor/customer ledgers.

05. EXPECTED OUTCOMES & ROI

- **100% Traceability:** Reverse trace any customer complaint to the exact machine, operator, and raw material batch in seconds.
- **Zero Double Entry:** Integrated billing, e-Invoicing, and E-Way bills directly from dispatched packing slips.
- **Live Executive Visibility:** Real-time overview of open orders, factory machine load, scrap rates, and payment receivables.
- **Error-Free Dispatches:** Barcode/QR verification guarantees zero wrong cartons loaded onto trucks.

06. PROJECT DELIVERABLES SUMMARY

Production ERP Core

ERPNext v15 + GST Compliance

Custom Next.js Web App

Dashboard, Scanner & Timeline

Dedicated VPS Setup

Dockerized Secure Cloud Deployment

HPOS Implementation Proposal | Page 2 of 2